KERALA POLYTECHNIC • SEMESTER 2 • COMMON COURSE

Environmental Science

A complete handbook for ecosystems, pollution control, renewable energy, solid-waste management, environmental law awareness, ISO 14000, carbon accounting and field-based case studies—with Malayalam support, worked problems, diagrams, revision tools and answer keys.

Course Code
2001

Semester
2

Credits
No Credit

Category
Common Course

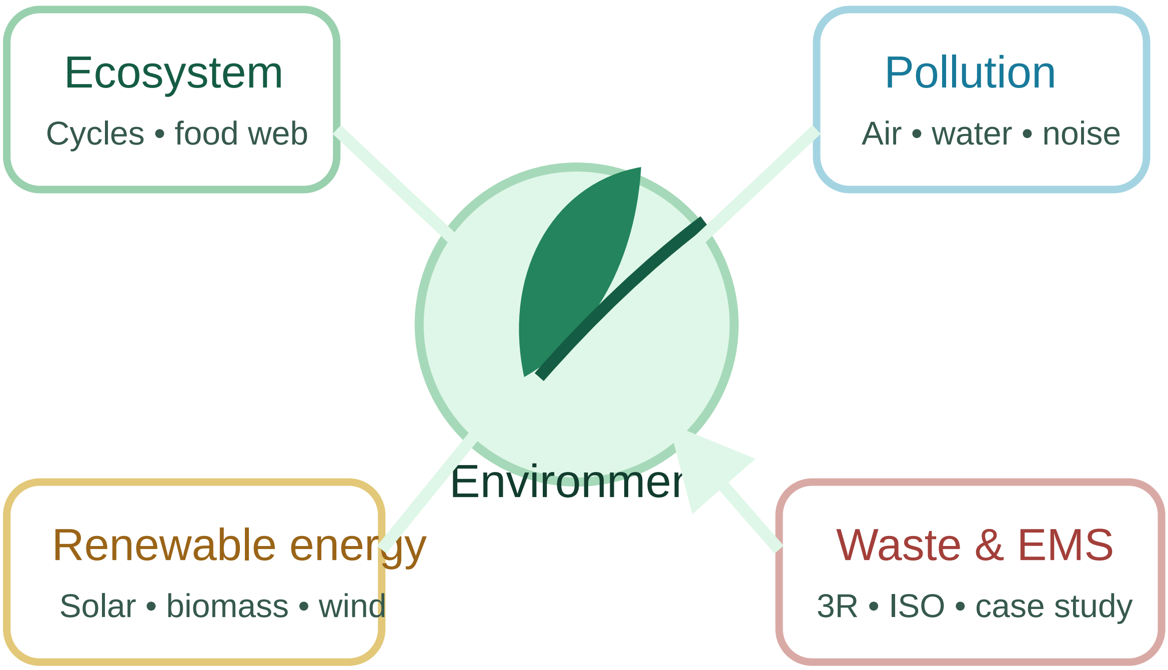
Periods/Week
3 (L:2 T:1 P:0)

Total Periods
45

Course Outcomes
4

Series Tests
2 × 1 hour

Understand systems. Measure impacts. Design better solutions.



Module 1 → Module 2 → Module 3 → Module 4 → Case study and sustainable engineering action

COURSE PURPOSE

Objectives and prerequisite

Understand ecosystems

Develop essential knowledge of ecosystem structure, energy transfer, nutrient cycles and global environmental processes.

Recognise environmental problems

Study the sources, effects and control of air, water, soil and noise pollution.

Apply technical thinking

Use engineering observation, measurement and design skills to propose workable environmental solutions.

Explore renewable energy

Understand solar, biomass, wind, hydrogen, ocean, tidal and geothermal resources and their harvesting processes.

Course 2001

Manage materials and waste

Study municipal, industrial, electronic, biomedical and hazardous waste, including 3R, recovery and disposal.

Prerequisite

Basic Chemistry from Semester 1 Applied Chemistry supports understanding of gases, water quality, reactions, fuels and materials.

പഠനരീതി: ഓരോ environmental topic-വും “source → pathway → effect → measurement → control → prevention” എന്ന ക്രമത്തിൽ പഠിച്ചാൽ ഓർക്കാൻ എളുപ്പമാണ്.

OFFICIAL OUTCOME STRUCTURE

Course outcomes and duration

CO	Outcome	Hours	Cognitive level
CO 1	Explain the ecosystem and terminology involved in it.	10	Understanding
CO 2	Explain air, water, soil and noise pollution, and control measures and acts.	11	Understanding
CO 3	Explain different renewable energy resources and efficient processes of harvesting.	9	Understanding
CO 4	Explain solid-waste management, ISO 14000 and environmental management, and conduct a case study on an environmental problem or sustainable-energy application.	13	Understanding

Hour verification: 10 + 11 + 9 + 13 = 43 teaching hours. Two one-hour series tests complete the 45-period semester.

CO–PO MAPPING

Outcome relationship

Course outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	—	—	—	—	—	—	—
CO2	—	—	—	—	2	—	—
CO3	—	—	—	—	3	—	—
CO4	—	—	—	—	—	3	3

In the syllabus legend, 3 means strongly mapped, 2 moderately mapped and 1 weakly mapped.

DETAILED SYLLABUS MAP

Module-by-module handbook structure

Module code	Official topic	Hours	Handbook expansion
M1.01	Structure of ecosystem; biotic and abiotic components	1	Levels of organisation, habitats, ecological pyramids, examples and system diagrams
M1.02	Food chain and food web	1	Trophic levels, energy transfer, grazing/detritus chains and stability
M1.03	Aquatic lentic/lotic and terrestrial ecosystem	1	Pond, lake, river, forest, grassland and desert comparison
M1.04	Carbon, nitrogen, sulphur and phosphorus cycles	4	Processes, reservoirs, human disturbances, diagrams and tracing problems
M1.05	Global warming, greenhouse effect, ozone depletion and acid rain	3	Mechanisms, effects, prevention and misconception correction
M2.01	Pollution/pollutant; natural and man-made air-pollution sources	3	Source–pathway–receptor model; refrigerants, IC engines and boilers
M2.02	Air pollutants and control equipment	3	Particulates, gases, bag filter, cyclone, ESP, absorber and catalytic converter
M2.03	Water pollution and pH, DO, BOD, COD	3	Parameters, interpretation, sampling awareness and solved comparisons
M2.04	Noise pollution and rules	2	Sources, decibel scale, effects, controls and logarithmic examples
Series Test I	Modules 1 and 2	1	Original practice test with complete answers
M3.01	Solar energy and collectors	2	Flat-plate collectors, coatings, pond, heater, dryer and still
M3.02	Biomass and biogas	2	Fuel properties, anaerobic digestion, storage and utilisation
M3.03	Wind energy	2	Resource, turbine parts, Indian context, benefits and challenges
M3.04	Hydrogen, ocean, tidal and geothermal energy	3	Principles, conversion routes, applications and comparison
M4.01	Municipal, e-waste and biomedical waste	2	Sources, characteristics, risks and segregation
M4.02	Metallic and non-metallic industrial waste	2	Lubricants, plastics, rubber, metals and recovery options
M4.03	Collection/disposal, 3R, recovery, landfill and hazardous waste	2	Waste hierarchy, logistics, landfill anatomy and treatment selection
M4.04	Syllabus-listed environmental acts and pollution-control boards	1	Purpose, institutional roles and current-source verification guidance
M4.05	Carbon credit and carbon footprint	1	Scopes, activity data, factors and solved footprint example
M4.06	Environmental management in fabrication industry	1	Aspect–impact register, controls, monitoring and improvement
M4.07	ISO 14000 implementation and benefits	1	EMS cycle, ISO 14001 concepts, documentation and audit
M4.08	Case study	3	Problem definition, data, engineering solution, sustainability and report
Series Test II	Modules 3 and 4	1	Original practice test with complete answers

HOW TO USE THIS HANDBOOK

Six-step learning workflow

1. Read the concept and identify all key terms.
2. Study the system diagram or process flow.
3. Follow one worked example or engineering scenario.

4. Answer the quick-check questions without looking back.
5. Attempt a case-based or numerical problem.
6. Use the revision, viva and master-answer sections for final preparation.

Answer-writing pattern: Definition → source/process → effect → control/prevention → labelled diagram or example → short conclusion.

Module 1 — Ecosystem and Environmental Processes

Understand ecosystem structure, energy transfer, aquatic and terrestrial systems, biogeochemical cycles and major atmospheric environmental problems.

M1.01 • 1 HOUR

Structure of an ecosystem

Abiotic components

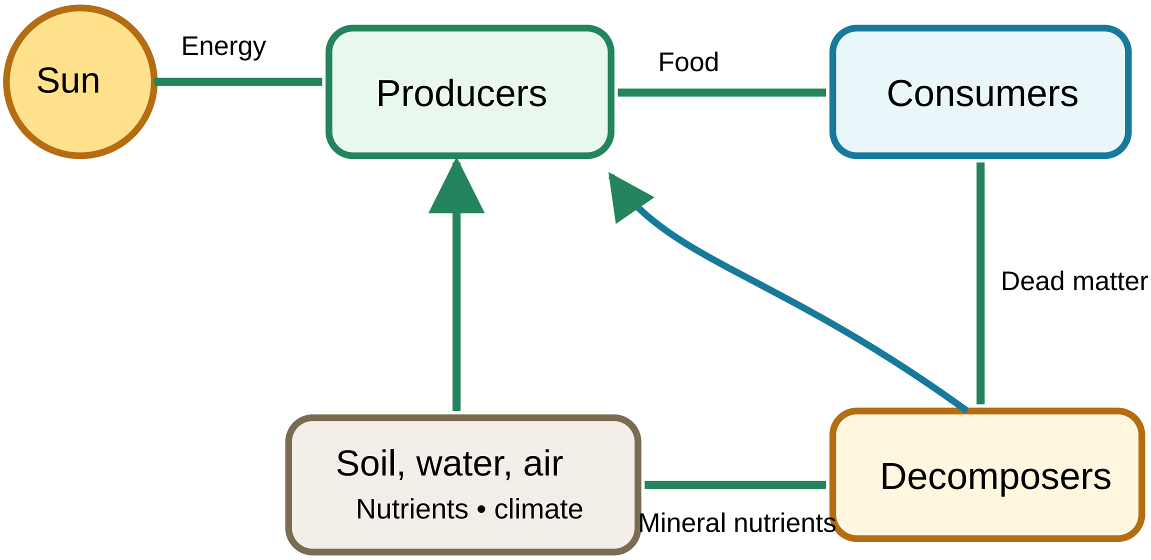
Non-living physical and chemical factors: sunlight, temperature, water, air, soil, minerals, pH, salinity and climate.

Biotic components

Living organisms grouped functionally as producers, consumers and decomposers.

System interaction

Energy enters mainly as sunlight, moves through organisms and leaves as heat, while matter cycles repeatedly.



Energy flows mainly in one direction; nutrients are recycled by decomposition and biogeochemical processes.

Functional groups

Group	Role	Examples	Engineering relevance
Producers	Convert solar or chemical energy into organic matter	Green plants, algae	Biomass production, oxygen release, carbon capture
Primary consumers	Feed directly on producers	Deer, rabbit, zooplankton	Transfer plant energy upward
Secondary/tertiary consumers	Feed on other consumers	Frog, snake, eagle	Control populations and maintain balance
Decomposers	Break down dead matter and wastes	Bacteria, fungi	Nutrient recycling, composting and treatment

Ecological organisation

Organism
One individual living being.

Population
Individuals of one species in an area.

Community
Interacting populations.

Ecosystem
Community plus abiotic environment.

Biome
Large climatic ecological region.

Biosphere
All ecosystems on Earth.

Ecological pyramids

Pyramid of numbers

Shows the number of organisms at each trophic level. It may be upright or inverted depending on the ecosystem.

Pyramid of biomass

Shows living mass at each trophic level. It can be inverted in some aquatic systems because phytoplankton renew rapidly.

Pyramid of energy

Shows energy transfer per unit area and time. It is always upright because energy is lost as heat at every transfer.

Worked Problem 1

Energy available across trophic levels

A grassland producer level captures 20,000 kJ. Assuming a simplified 10% transfer at each level, estimate energy available to primary, secondary and tertiary consumers.

$$\text{Primary} = 20{,}000 \times 0.10 = 2{,}000\ \text{kJ}$$

$$\text{Secondary} = 2{,}000 \times 0.10 = 200\ \text{kJ}$$

$$\text{Tertiary} = 200 \times 0.10 = 20\ \text{kJ}$$

Interpretation: Large producer energy supports much less energy at the top. This limits the number and biomass of top predators.

> ഓരോ trophic level-ലും വലിയൊരു ഭാഗം energy heat ആയി നഷ്ടപ്പെടുന്നതിനാൽ energy pyramid എല്ലായ്പ്പോഴും upright ആണ്.

M1.02 • 1 HOUR

Food chain and food web

Food chain

A linear feeding sequence showing transfer of matter and energy from one organism to another.

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Grass → Grasshopper → Frog → Snake → Eagle
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Arrow meaning: “is eaten by” or direction of energy transfer.
Course 2001

Food web

A network of interconnected food chains. It better represents real ecosystems because organisms usually have multiple food sources and predators.

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Grass → Rabbit → Fox
Grass → Grasshopper → Frog → Snake
Seeds → Mouse → Snake → Eagle
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Types of food chain

Type	Starts with	Example	Importance
Grazing food chain	Living green plant	Grass → deer → tiger	Transfers freshly fixed solar energy
Detritus food chain	Dead organic matter	Leaf litter → earthworm → bird	Recycles wastes and dead biomass

Why food webs improve stability: If one food source declines, some organisms may shift to alternatives. However, highly specialised species may still be vulnerable.

Scenario

Removing a predator

If snakes are heavily removed from a crop-field food web, rodent numbers can rise. More rodents may damage crops and consume stored grain. The problem is not only “one species missing”; it is a cascade through several links.

Engineering lesson: Pest management should consider ecosystem interactions rather than relying only on repeated chemical control.

M1.03 • 1 HOUR

Aquatic and terrestrial ecosystems

Ecosystem	Key condition	Typical producers	Typical consumers	Major environmental pressure
Lentic	Standing water: pond, lake, reservoir	Phytoplankton, algae, aquatic plants	Zooplankton, fish, insects	Eutrophication, sedimentation, oxygen depletion
Lotic	Flowing water: stream, river	Attached algae, riparian vegetation	Insects, fish, amphibians	Flow alteration, sewage, sand mining
Forest	Tree-dominated terrestrial system	Trees and understory plants	Insects, birds, mammals	Deforestation and fragmentation
Grassland	Grass-dominated with periodic grazing/fire	Grasses and herbs	Grazers and predators	Overgrazing and conversion
Desert	Very low rainfall and high stress	Xerophytes and seasonal plants	Reptiles, insects, adapted mammals	Water extraction and habitat disturbance

Lentic versus lotic: Lentic systems often develop vertical zones and may stratify. Lotic systems are shaped by flow velocity, channel form, oxygen exchange and upstream activities.

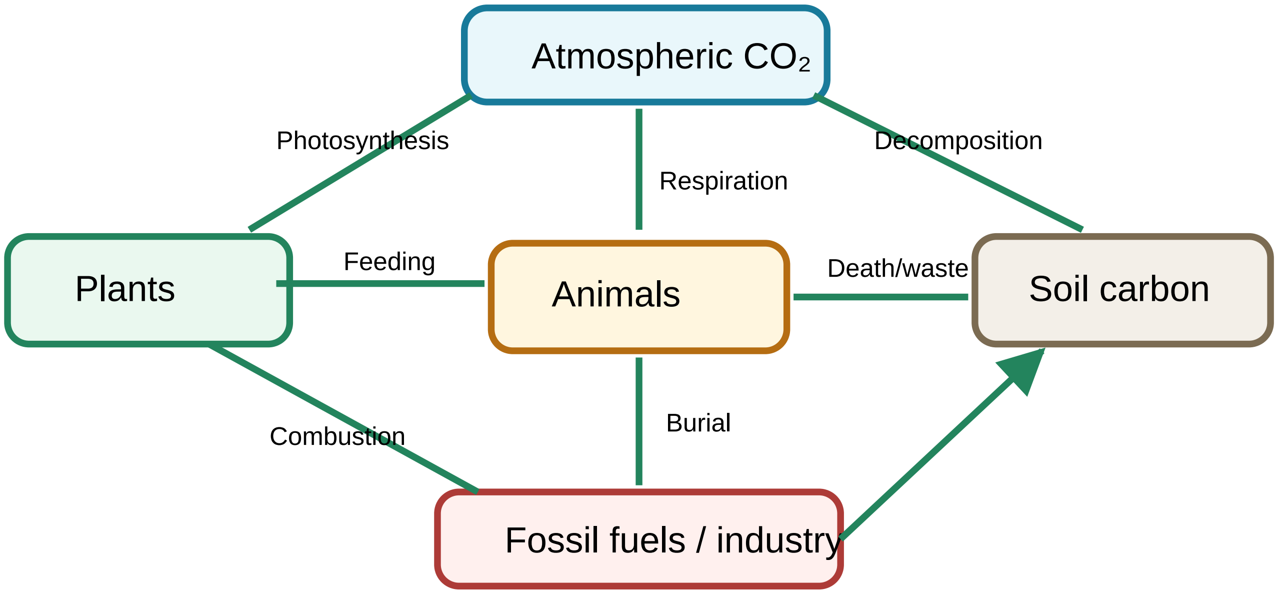
ഓർക്കുക: Lentic = standing water; Lotic = flowing water. Pond/lake lentic ആണ്; stream/river lotic ആണ്.

M1.04 • 4 HOURS

Biogeochemical cycles

Biogeochemical cycles move essential elements among atmosphere, water, soil, rocks and organisms. Human activity can change the speed, location and form of these transfers.

Carbon cycle



Photosynthesis removes atmospheric CO_2; respiration, decomposition and combustion return it.

Process	Direction	Human influence
Photosynthesis	Atmosphere → biomass	Deforestation reduces uptake
Respiration	Biomass → atmosphere	Natural biological process
Decomposition	Dead matter → soil/atmosphere	Landfill conditions may produce methane
Combustion	Fuel/biomass → atmosphere	Fossil-fuel burning increases rapid release
Ocean exchange	Atmosphere ↔ ocean	Warming and acidification alter balance

Nitrogen cycle

Nitrogen fixation

Converts atmospheric N_2 into biologically usable forms through microbes, lightning or industrial processes.

Nitrification

Microbial conversion of ammonia/ammonium into nitrite and nitrate.
Course 2001

Assimilation

Plants absorb nitrate/ammonium and build proteins; animals obtain nitrogen by feeding.

Ammonification

Decomposers convert organic nitrogen in wastes and dead matter into ammonia/ammonium.

Denitrification

Microbes convert nitrate back to gaseous forms under low-oxygen conditions.

Human disturbance

Excess fertiliser and sewage can cause nutrient enrichment, algal blooms and nitrate contamination.

Sulphur cycle

Sulphur occurs in rocks, soil, water, living matter and atmospheric gases. Weathering releases sulphates; plants assimilate them; food transfers sulphur; decomposition returns it. Volcanic emissions and combustion of sulphur-containing fuels release SO_2, which can form acids in the atmosphere.

Phosphorus cycle

Phosphorus has no major gaseous phase under normal conditions. Weathering releases phosphate from rocks, plants absorb it, animals obtain it by feeding, and decomposition returns it to soil and water. Runoff carrying fertiliser or detergent phosphate can intensify eutrophication.

Cycle	Main reservoir	Important processes	Typical disturbance
Carbon	Atmosphere, ocean, rocks, biomass	Photosynthesis, respiration, combustion	Fossil-fuel use and deforestation
Nitrogen	Atmosphere	Fixation, nitrification, assimilation, denitrification	Excess fertiliser and reactive nitrogen emissions
Sulphur	Rocks, sediments and ocean	Weathering, assimilation, decomposition, oxidation	Fuel combustion and mining
Phosphorus	Rocks and sediments	Weathering, uptake, food transfer, sedimentation	Runoff and eutrophication

Worked Problem 2
Tracing fertiliser nitrogen

A field receives 100 kg of nitrogen. Suppose 55 kg is taken up by crops, 15 kg remains in soil, 10 kg is lost to the atmosphere and the rest leaves through runoff/leaching.

$$\text{Runoff/leaching} = 100 - (55 + 15 + 10) = 20\ \text{kg}$$

Interpretation: The 20 kg loss can affect groundwater or surface water. Better timing, dose control, soil testing and buffer strips can reduce loss.

M1.05 • 3 HOURS

Global warming, greenhouse effect, ozone depletion and acid rain

Natural greenhouse effect and enhanced warming

The atmosphere allows much incoming shortwave solar radiation to reach Earth. The warmed surface emits longwave infrared radiation. Greenhouse gases absorb and re-emit part of this outgoing energy, keeping the planet warmer than it would otherwise be. Human activities raise concentrations of several greenhouse gases, strengthening this effect.

Gas	Major human-related sources	Environmental relevance
Carbon dioxide	Fossil-fuel combustion, cement production, land-use change	Major long-lived driver of warming
Methane	Oil/gas systems, livestock, rice, landfill, wetlands	Strong warming influence and ozone precursor
Nitrous oxide	Fertiliser, manure, industry and combustion	Long-lived greenhouse gas and ozone-depleting influence
Fluorinated gases	Refrigeration, electronics and industry	Some have very high warming potential

Causes

- Combustion of coal, oil and gas
- Deforestation and land-use change
- Industrial processes and cement
- Agriculture and livestock
- Waste decomposition and refrigerant leakage

Effects

- Rising average temperature and heat extremes
- Changes in rainfall and drought/flood risk
- Ice loss and sea-level rise
- Ecosystem and crop stress
- Health, infrastructure and livelihood impacts

Common misconception: The greenhouse effect is not the same as ozone depletion. They involve different gases, mechanisms and consequences, although some substances can influence both.

Ozone depletion

Stratospheric ozone absorbs harmful ultraviolet radiation. Certain long-lived chlorine- and bromine-containing compounds can reach the stratosphere, where radiation releases reactive atoms that catalytically destroy ozone. Consequences include increased UV exposure, skin and eye damage, and effects on crops and plankton.

Acid rain

Sulphur dioxide and nitrogen oxides can react in the atmosphere to form sulphuric and nitric acids. Wet deposition occurs in rain, fog or snow; dry deposition occurs as gases and particles. Effects include acidification, nutrient loss from soil, aquatic stress and corrosion of structures.

Problem	Main mechanism	Control direction
Enhanced global warming	Higher greenhouse-gas concentration traps more outgoing heat	Efficiency, renewable energy, low-carbon transport, methane control, carbon sinks
Ozone depletion	Catalytic destruction of stratospheric ozone	Phase-out controlled substances, recover refrigerants, prevent leakage
Acid rain	SO_2 and NO_x form acids in atmosphere	Cleaner fuels, flue-gas control, low-NO_x combustion, transport control

Worked Problem 3 Simple emissions comparison

Option A consumes 1,000 kWh from a source with an illustrative emission factor of 0.70 kg CO_2e/kWh. Option B uses 650 kWh at the same factor.

$$A = 1{,}000 \times 0.70 = 700 \text{ kg } CO_2e$$

$$B = 650 \times 0.70 = 455 \text{ kg } CO_2e$$

$$\text{Illustrative reduction} = 700 - 455 = 245 \text{ kg } CO_2e$$

Important: Emission factors vary by place, time and method. Use the current authorised factor for real reporting.

Module 1 quick check

1. Why is energy flow unidirectional while nutrients cycle?

Energy is progressively dissipated as heat, whereas atoms are conserved and reused through biological, geological and chemical processes.

2. Differentiate a food chain and food web.

A food chain is one linear feeding path; a food web is a network of interconnected feeding paths.

3. Why can excess phosphorus cause eutrophication?

It can remove a nutrient limitation, encouraging algal growth. Decomposition of excess biomass consumes oxygen.

4. Greenhouse effect versus ozone depletion?

The greenhouse effect concerns heat retention by infrared-absorbing gases; ozone depletion concerns loss of stratospheric UV protection.

Module 2 — Pollution, Measurement and Control

Explain sources, effects and control of air, water, soil and noise pollution, including equipment, indicators and syllabus-listed regulatory awareness.

M2.01 • 3 HOURS

Pollution, pollutants and sources

Pollution

An undesirable change in the physical, chemical or biological quality of the environment that harms living organisms, materials, comfort or useful environmental functions.

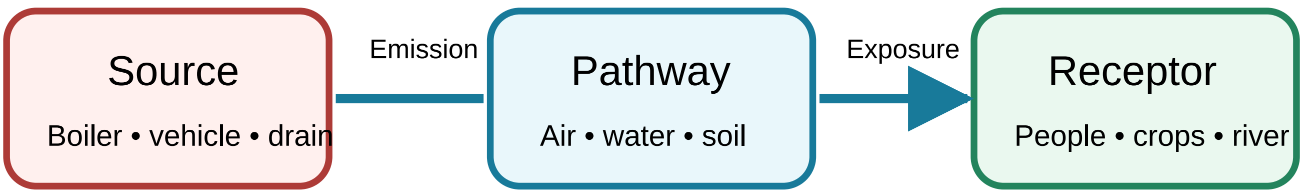
Pollutant

A substance, form of energy or biological agent present at a concentration, duration or location capable of causing adverse effects.

Receptor

The person, organism, ecosystem, material or process that receives exposure and may be affected.

Source–pathway–receptor model



Control can act at the source, interrupt the pathway or protect the receptor; source prevention is usually preferred.

Natural and human-made sources

Source class	Examples	Typical pollutants	Control approach
Natural	Dust storms, wildfire, sea spray, volcanic activity, pollen	Particles, gases, aerosols, biological matter	Monitoring, warning, exposure reduction and land management
Transport	Internal-combustion vehicles	CO, NO_x, hydrocarbons, particles, CO_2	Maintenance, cleaner fuels, catalytic control, public transport, electrification
Boilers/furnaces	Industrial and commercial combustion	Particles, SO_2, NO_x, CO and ash	Fuel quality, combustion optimisation and flue-gas treatment
Refrigeration	Leaks during operation or servicing	Refrigerants with ozone or climate impacts	Leak prevention, recovery, correct charging and approved alternatives
Construction	Excavation, crushing and transport	Dust and diesel exhaust	Water suppression, covers, housekeeping and equipment maintenance
Agriculture	Fertiliser, livestock, residue burning	Ammonia, methane, nitrous oxide, smoke	Nutrient planning, manure management and no-burn alternatives

Primary and secondary pollutants

Primary pollutants are emitted directly, such as CO, SO_2 or dust. **Secondary pollutants** form in the environment through reactions, such as ground-level ozone and some secondary particles.

> **ഓർക്കുക:** Source-ൽ നേരിട്ട് പുറപ്പെടുന്നത് primary pollutant; atmosphere-ൽ reaction കഴിഞ്ഞ് രൂപപ്പെടുന്നത് secondary pollutant.

M2.02 • 3 HOURS

Air pollutants, effects and control equipment

Particulate and gaseous pollutants

Pollutant	Typical source	Effect	Suitable control direction
Coarse/fine particles	Combustion, grinding, road dust	Respiratory/cardiovascular stress, visibility and soiling	Cyclone, bag filter, ESP, enclosure and suppression
SO_2	Sulphur-containing fuel	Respiratory irritation, acid deposition and corrosion	Low-sulphur fuel, desulphurisation and absorption
NO_x	High-temperature combustion	Ozone/smog formation and respiratory impacts	Combustion control, catalytic treatment
CO	Incomplete combustion	Reduces oxygen transport in blood	Complete combustion, ventilation and catalytic oxidation
VOCs/hydrocarbons	Solvents, fuel and incomplete combustion	Odour, toxicity and ozone formation	Substitution, capture, adsorption, oxidation
Ground-level ozone	Secondary photochemical formation	Lung and crop damage	Control NO_x and VOC precursors

Particle-control devices

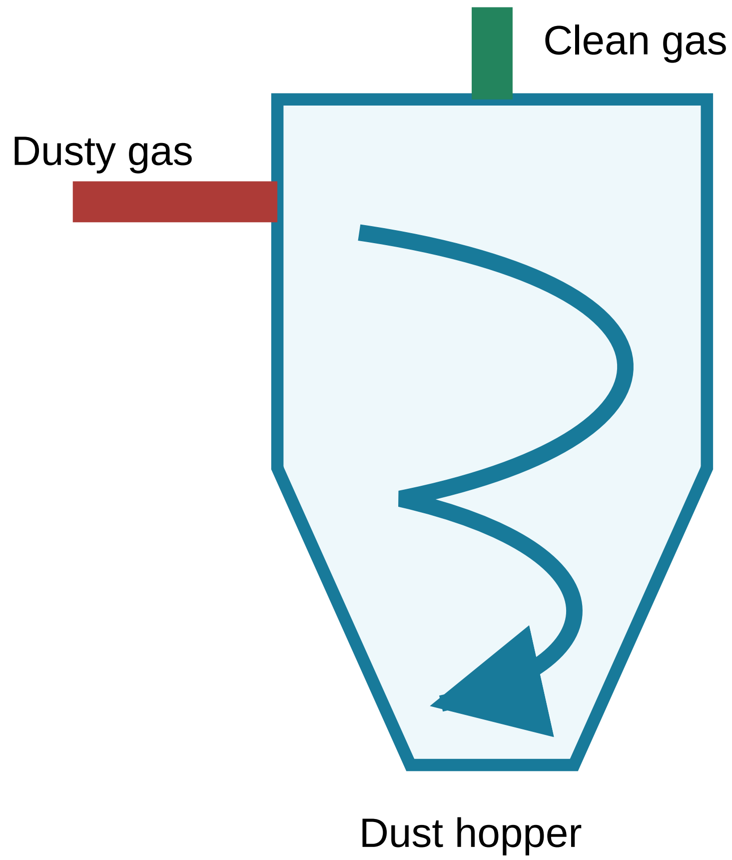
Control Device 1

Cyclone separator

Dust-laden gas enters tangentially and forms a vortex. Centrifugal action drives heavier particles toward the wall; particles lose momentum and fall into a hopper while cleaner gas exits.

Strengths: simple, robust, low pressure loss, suitable as pre-cleaner.

Limitations: less efficient for very fine particles.



Control Device 2

Bag filter

Gas passes through fabric bags. Particles are captured on fibres and in the dust cake. Periodic shaking, reverse air or pulse cleaning removes dust into a hopper.

Best suited

High removal of many dry particles, including fine dust.

Watch for

Temperature, moisture, corrosiveness and fire/explosion risk.

Maintenance

Bag integrity, differential pressure and dust disposal.

Control Device 3

Electrostatic precipitator (ESP)

Particles are electrically charged and attracted to oppositely charged collection plates. Rapping dislodges collected material into hoppers.

Strengths: high efficiency for large gas flows with relatively low pressure drop. **Limitations:** performance depends on particle electrical properties, installation and operating conditions.

Gaseous-pollution control

Absorber / scrubber

Transfers a gaseous pollutant from gas into a liquid. Selection depends on solubility and reaction chemistry. A packed tower provides large contact area.

Example: alkaline liquid absorbing an acidic gas.

Catalytic converter

Uses catalysts to promote oxidation and reduction reactions in vehicle exhaust, reducing CO, unburned hydrocarbons and NO_x when the engine and fuel-control system operate correctly.

Important: It does not eliminate CO_2 produced from fuel combustion.

Control-device selection

Situation	Likely first choice	Reason
Large, coarse, dry particles before another unit	Cyclone	Robust pre-cleaning
Fine dry dust with suitable temperature	Bag filter	High filtration efficiency
Very large flue-gas flow	ESP	Large capacity and low pressure drop
Soluble/reactive gas	Absorber	Gas-to-liquid mass transfer
Vehicle CO/HC/NO_x	Catalytic converter	Catalysed exhaust reactions

Worked Problem 4 — Collection efficiency and outlet concentration

A dust-control unit receives gas containing 500 mg/m³ of particulate matter and removes 92%.

$$\text{Removed concentration} = 500 \times 0.92 = 460\ \text{mg/m}^3$$

$$\text{Outlet concentration} = 500 - 460 = 40\ \text{mg/m}^3$$

Check: 40/500 = 0.08, so 8% remains and 92% is removed.

Air-quality standards: Standards set concentration limits or assessment methods for specified pollutants and averaging periods. Real compliance work must use the current official standard, approved sampling method and calibrated instruments.

M2.03 • 3 HOURS

Water pollution and quality parameters

Sources and effects

Source	Possible contaminants	Effect	Prevention/control
Domestic sewage	Organic matter, nutrients, pathogens	Oxygen depletion and disease risk	Collection, treatment and sanitation
Industry	Acids/alkalis, metals, solvents, heat and organics	Toxicity, pH change and ecosystem damage	Cleaner production and effluent treatment
Agricultural runoff	Nutrients, sediment and pesticides	Eutrophication and contamination	Correct application, buffers and erosion control
Urban runoff	Oil, litter, metals and sediment	Short pollution pulses and flooding	Drain maintenance and stormwater management
Solid-waste dumping	Leachate and plastics	Ground/surface-water pollution	Segregation and engineered disposal

Important parameters

pH

A logarithmic measure related to hydrogen-ion activity. pH below 7 is acidic, 7 neutral under standard classroom simplification, and above 7 alkaline.

Dissolved oxygen (DO)

Oxygen dissolved in water and available to aquatic organisms. Low DO may indicate high oxygen demand, poor mixing or warm/stagnant conditions.

Biochemical oxygen demand (BOD)

Oxygen consumed by microorganisms while decomposing biodegradable organic matter under specified test conditions.

Chemical oxygen demand (COD)

Oxygen-equivalent measure of substances chemically oxidised by a strong reagent under the test method.

Turbidity

Cloudiness caused by suspended and colloidal particles, affecting light penetration and treatment.

Nutrients and solids

Nitrogen, phosphorus, dissolved solids and suspended solids influence eutrophication, salinity and treatment needs.

Pattern	Likely interpretation
High BOD + low DO	Strong biodegradable organic load may be consuming oxygen
COD much higher than BOD	Substantial chemically oxidisable material may be poorly biodegradable
High nutrients + algal bloom	Eutrophication risk
Extreme pH	Possible industrial/chemical input and toxicity
High turbidity	Sediment, runoff, disturbance or treatment problem

Worked Problem 5 Interpreting two water samples

Sample	DO (mg/L)	BOD (mg/L)	COD (mg/L)
A	7.2	2	12
B	2.1	28	75

Interpretation: Sample B shows much higher oxygen demand and much lower dissolved oxygen, indicating a substantially greater pollution load than A. COD exceeds BOD in both because the chemical method can measure a wider range of oxidisable substances.

> **Do not diagnose from one number alone:** Sampling location, temperature, test method, time and other parameters must be considered.

Worked Problem 6 Simple BOD removal efficiency

An influent has BOD 180 mg/L and treated effluent has BOD 27 mg/L.

$$\text{Removal efficiency} = (180 - 27) / 180 \times 100 = 85\%$$

The treatment removed 153 mg/L BOD-equivalent load from the measured concentration basis.

Soil pollution awareness

Although the detailed outline emphasises air, water and noise, the course outcome also names soil pollution. Soil can be contaminated by uncontrolled dumping, oil, heavy metals, pesticides, salts and industrial residues. Effects include reduced fertility, food-chain transfer, groundwater contamination and direct exposure.

> **Parameter memory:** DO കൂടുതലാകുന്നത് പൊതുവെ നല്ല oxygen availability സൂചിപ്പിക്കും; BOD/COD ഉയരുന്നത് pollution load കൂടുതലാകാമെന്ന് സൂചിപ്പിക്കും.

M2.04 • 2 HOURS

Noise pollution, measurement and control

Noise is unwanted or harmful sound. Its impact depends on sound level, frequency, duration, timing, impulsiveness, distance and receiver sensitivity.

Sources

Traffic, aircraft, construction, machinery, generators, workshops, loudspeakers and household equipment.
Course 2001

Effects

Hearing damage, sleep disturbance, stress, reduced concentration, communication interference and wildlife disturbance.

Control hierarchy

Quiet source/equipment → enclosure or barrier → distance/layout → maintenance → scheduling → hearing protection.

Decibel scale

Sound-pressure level is logarithmic. Therefore, two equal sound sources do not add arithmetically in decibels.

$$L_t = 10 \log_{10}(10\text{^}(L_1/10) + 10\text{^}(L_2/10) + \ldots)$$

Worked Problem 7

Combining two equal noise sources

Two independent machines each produce 80 dB at a point.

$$L_t = 10 \log_{10}(10^8 + 10^8) = 10 \log_{10}(2 \times 10^8)$$

$$L_t \approx 83 \text{ dB}$$

Two equal sources increase the level by about 3 dB, not to 160 dB.

Worked Problem 8

Effect of distance in a simplified free field

For a point source in an ideal free field, doubling distance reduces sound level by approximately 6 dB. If a machine produces 92 dB at 2 m, an approximate level at 4 m is 86 dB and at 8 m is 80 dB. Real rooms, reflections and multiple sources alter this estimate.

Control at source, path and receiver

Location	Examples
Source	Quieter equipment, balancing, lubrication, reduced speed, vibration isolation
Path	Enclosures, acoustic barriers, absorptive lining, increased distance
Receiver	Cabins, work rotation, limiting exposure, suitable hearing protection

Regulatory awareness: The syllabus lists the Noise Pollution (Regulation and Control) Rules, 2000. For actual compliance, use the current official text, amendments, zone classification, measurement protocol and competent-authority guidance.

SERIES TEST I • 1 HOUR

Original practice test — Modules 1 and 2

This is a syllabus-based practice test, not an official board question paper.

1. Define ecosystem and distinguish biotic and abiotic components.
2. Explain food chain, food web and ecological pyramid.
3. Draw and explain any one biogeochemical cycle.
4. Differentiate greenhouse effect, ozone depletion and acid rain.
5. Define pollution and pollutant using the source–pathway–receptor model.
6. Explain working, advantage and limitation of a cyclone or bag filter.
7. Differentiate DO, BOD and COD.
8. A treatment plant reduces BOD from 240 mg/L to 36 mg/L. Calculate percentage removal.
9. Explain effects and control of noise pollution.
10. Two equal sources produce 85 dB each. Estimate the combined level.

Module 3 — Renewable Energy Resources and Harvesting

Solar, biomass, wind, hydrogen, ocean, tidal and geothermal energy: principles, equipment, applications, calculations, benefits and limitations.

M3.01 • 2 HOURS

Solar energy

Solar systems capture radiation, convert it into heat or electricity, transfer or store it and deliver it to a load. Output depends on location, season, time, cloud, orientation, tilt, shading and losses.

Direct radiation

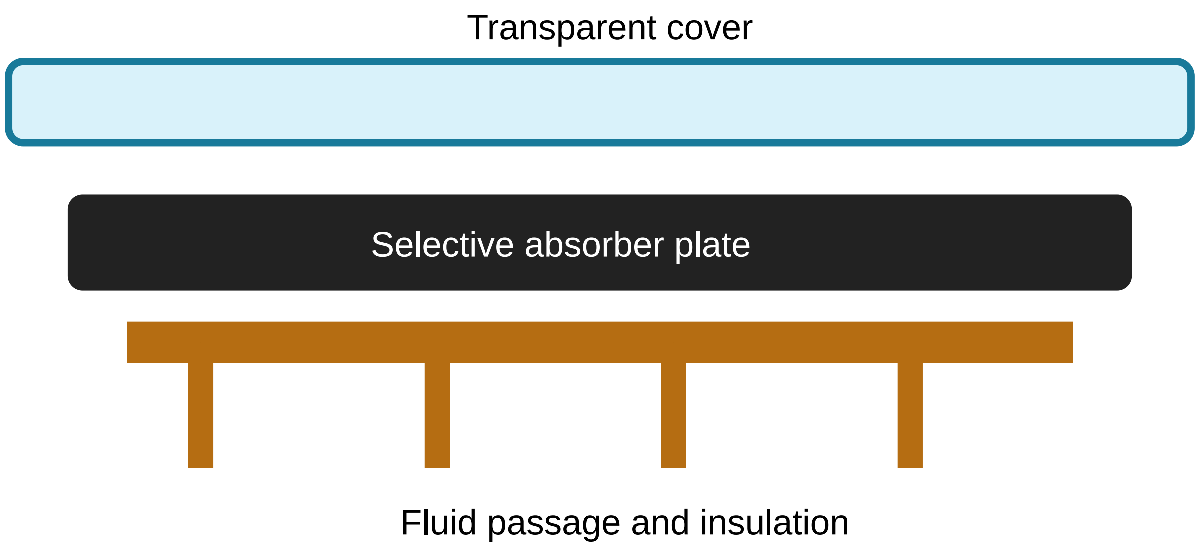
Reaches a surface without atmospheric scattering; concentrating collectors mainly require it.

Diffuse radiation

Scattered by air, cloud and particles; flat-plate collectors can receive it.

Global radiation

Total direct and diffuse radiation received by the surface.



Cover reduces convective loss; the dark absorber converts radiation to heat; tubes or air channels carry heat; insulation reduces rear loss.

System	Principle	Application	Limitation
Liquid flat plate	Water/antifreeze flows through tubes on absorber	Water and low-temperature process heating	Scaling, freezing and leakage
Air flat plate	Air passes over/under heated absorber	Space and crop drying	Lower heat capacity and duct losses
Solar water heater	Collector transfers heat to stored water	Domestic/institutional hot water	Cloud and backup requirement
Solar dryer	Solar-heated air removes moisture	Food/crop preservation	Temperature-humidity control
Solar still	Evaporation followed by condensation	Small-scale distilled water	Low daily output
Solar pond	Salinity gradient suppresses convection and stores heat	Low-grade heat	Area and gradient maintenance
Advanced collector	Concentrates direct radiation	Higher-temperature heat	Tracking and clear-sky dependence

Worked Problem 9 Solar water-heating energy

Raise 100 L of water from 25°C to 60°C. Take density 1 kg/L and specific heat 4.18 kJ/kg·°C.

$$Q = m\,c\,\Delta T = 100 \times 4.18 \times 35 = 14{,}630\ \text{kJ} = 14.63\ \text{MJ}$$

At 50% overall efficiency, simplified incident energy required = 14.63/0.50 = 29.26 MJ.

Worked Problem 10 Collector output

Area 4 m², irradiance 700 W/m² and efficiency 45%.

$$\text{Useful power} = 4 \times 700 \times 0.45 = 1{,}260\ \text{W} = 1.26\ \text{kW}$$

For five equivalent full-sun hours, simplified useful energy = 6.3 kWh.
Course 2001

Solar memory: Coating radiation absorb ചെയ്യുകയും thermal radiation loss കുറയ്ക്കുകയും ചെയ്യുന്നു.

M3.02 • 2 HOURS

Biomass and biogas

Biomass is renewable organic material from plants, animals and biodegradable residues. Moisture, ash, volatile matter, fixed carbon, calorific value and bulk density affect fuel behaviour.

Anaerobic digestion stages

1. **Hydrolysis:** complex material becomes soluble molecules.
2. **Acidogenesis:** products become organic acids, alcohols, H_2 and CO_2.
3. **Acetogenesis:** intermediates become acetate, H_2 and CO_2.
4. **Methanogenesis:** microorganisms produce methane-rich biogas.

Biogas use

Cooking, heating, engine-generator fuel after cleaning, or upgrading to biomethane.

Digestate

May return nutrients when pathogens and contaminants are properly managed.

Key controls

Temperature, pH, loading, retention time, mixing, toxicity, gas tightness and safety.

Worked Problem 11

Estimated gas yield

Feed = 20 kg/day; illustrative yield = 0.04 m³/kg.

$$\text{Biogas} = 20 \times 0.04 = 0.80 \text{ m}^3/\text{day}$$

At 60% methane, methane volume = 0.48 m³/day. Actual yield depends on substrate and operating condition.

Safety: Biogas is flammable and may contain toxic/corrosive H_2S. Provide ventilation, leak control, pressure relief and trained operation.

M3.03 • 2 HOURS

Wind energy

A wind turbine converts air-stream kinetic energy into rotor motion and electricity. Main parts are blades, hub, nacelle/drivetrain, generator, controls, tower and foundation.

$$\text{Available wind power} = \tfrac{1}{2}\,\rho\,A\,v^3$$

Actual output also includes the power coefficient and mechanical/electrical efficiencies. Because power varies with the cube of wind speed, site measurement is essential.

Worked Problem 12

Approximate turbine output

$\rho = 1.225$ kg/m^3, $A = 50$ m^2, $v = 8$ m/s and power coefficient = 0.35.

$$P = \tfrac{1}{2} \times 1.225 \times 50 \times 8^3 \times 0.35 \approx 5{,}488\ \text{W} = 5.49\ \text{kW}$$

Benefits	Challenges and controls
No fuel combustion during operation	Variable output needs grid/storage planning
Low operational water use	Visual, acoustic and shadow issues need siting
Land may retain agricultural use	Wildlife risk needs assessment and mitigation
Scalable technology	Roads, foundations and maintenance have impacts

ഓർക്കുക: Wind speed ചെറിയ തോതിൽ മാറിയാലും power വലിയ തോതിൽ മാറാം, കാരണം v^3 ബന്ധമാണ്.

M3.04 • 3 HOURS

Hydrogen, ocean, tidal and geothermal energy

Source	Origin/conversion	Application	Main challenge
Hydrogen	Produced by electrolysis, reforming or other routes; used in fuel cells/combustion	Industry, transport and storage	Production emissions, efficiency, cost, storage and safety
Wave	Wind-generated wave motion drives a device	Coastal electricity	Storm survival and maintenance
Tidal range	Sea-level difference drives barrage/lagoon turbines	Predictable electricity	Ecology and sediment change
Tidal stream	Underwater current drives turbine	Predictable marine power	Installation and marine-use conflict
Ocean thermal	Surface/deep-water temperature difference drives heat engine	Power and possible co-products	Low efficiency and large seawater flow
Geothermal	Earth heat supplies steam/hot water	Power and direct heating	Location, drilling, scaling and induced effects

Geothermal plant concepts

Dry steam

Natural steam drives a turbine directly.

Flash steam

Hot pressurised water is depressurised so part flashes to steam.

Binary cycle

Geothermal fluid heats a separate low-boiling working fluid.

Worked Problem 13

Electrolyser electricity

An illustrative electrolyser uses 52 kWh per kg hydrogen. For 100 kg:

$$\text{Electricity} = 52 \times 100 = 5{,}200\ \text{kWh}$$

Life-cycle emissions depend strongly on the electricity source.

Selection principle: Match resource to load, location, scale, skill, cost, reliability, maintenance and environmental safeguards. A hybrid solution may be better than one technology.

Module 3 quick check

Why is selective coating important?

It combines strong solar absorption with reduced thermal radiation loss.

Name four anaerobic-digestion stages.

Hydrolysis, acidogenesis, acetogenesis and methanogenesis.

Why measure wind at the site?

Wind varies with location, height, terrain and time, and power is highly sensitive to speed.

Is hydrogen a primary source?

It is generally an energy carrier; energy is required to produce it.

Module 4 — Solid Waste, Environmental Management and ISO 14000

Explain waste sources and characteristics, segregation, collection, recovery, disposal, carbon concepts, institutional roles, fabrication-industry management and environmental management systems.

M4.01 • 2 HOURS

Municipal solid waste, e-waste and biomedical waste

Waste management begins with correct identification and segregation. Mixing incompatible or hazardous material with ordinary waste increases exposure, treatment difficulty and cost.

Waste stream	Typical source	Characteristics	Main risk	Preferred management direction
Municipal solid waste	Homes, shops, institutions and streets	Food waste, paper, plastics, glass, metals, inert matter	Odour, vectors, leachate, litter and methane	Source segregation, collection, composting/recycling, recovery and engineered disposal
E-waste	Discarded electrical/electronic equipment	Metals, plastics, glass, circuit boards, batteries	Lead, mercury and other hazardous constituents; unsafe informal processing	Authorised collection, repair/reuse, producer systems and controlled recycling
Biomedical waste	Healthcare diagnosis, treatment and research	Infectious waste, sharps, contaminated material, pharmaceuticals	Infection, injury and chemical exposure	Segregation at source, labelled containers, trained handling and authorised treatment

Important waste characteristics

Physical

Density, moisture, particle size, composition and compactability affect collection and processing.

Chemical

Calorific value, ash, carbon, nutrients, metals, pH and reactivity affect recovery and treatment.

Biological

Biodegradability, pathogens and putrescibility influence composting, odour and health risk.

Segregation principle: Sharps, batteries, chemicals, infectious waste and ordinary recyclables must not be mixed. Real handling must follow current local and sector-specific rules.

Worked Problem 14 Daily municipal-waste estimate

A campus has 800 users and an observed generation rate of 0.35 kg/person·day.

$$\text{Daily waste} = 800 \times 0.35 = 280 \text{ kg/day}$$

If 45% is biodegradable, biodegradable fraction = 280 × 0.45 = 126 kg/day. This estimate helps size segregation, collection and treatment systems.

ഓർക്കുക: Waste treatment-ന് മുമ്പ് source segregation നിർണായകം. Mixed waste ആയാൽ recycling/composting quality കുറഞ്ഞുപോകും.

M4.02 • 2 HOURS

Metallic and non-metallic industrial wastes

Waste	Fabrication/industrial source	Concern	Management option
Metal offcuts and chips	Cutting, machining and punching	Sharp edges, oil contamination and loss of valuable metal	Segregate by alloy, drain fluids, reuse/recycle
Grinding dust	Grinding and finishing	Fine particles and possible metal exposure	Local extraction, sealed collection and authorised recovery/disposal
Spent lubricants	Machines, gearboxes and maintenance	Soil/water contamination and fire risk	Prevent leaks, labelled storage and authorised recovery
Plastic scrap	Packaging, moulding and components	Persistence, contamination and mixed polymer streams	Reduction, clean segregation and suitable recycling
Rubber waste	Seals, belts, tyres and offcuts	Fire, storage and slow degradation	Retreading/reuse, material recovery or authorised treatment
Paint/sludge/solvent residues	Surface preparation and coating	Flammability, VOCs and hazardous constituents	Substitution, process control, closed containers and authorised disposal

Material-flow thinking

For each input, track quantity purchased, quantity in product, reusable return, recyclable scrap, emissions and disposal. A material balance can reveal avoidable loss.

Worked Problem 15 Metal utilisation

A workshop purchases 1,000 kg sheet metal. Product contains 780 kg, clean segregated offcuts are 170 kg, and mixed/unaccounted loss is 50 kg.

$$\text{Product yield} = 780 / 1{,}000 \times 100 = 78\%$$

$$\text{Recoverable offcut rate} = 170 / 1{,}000 \times 100 = 17\%$$

Loss requiring investigation = 5%

Improvement may include nesting optimisation, better segregation, chip recovery, purchase-size selection and accurate weighing.

M4.03 • 2 HOURS

Collection, 3R, energy recovery, landfill and hazardous waste

3R principles

Reduce

Avoid generating waste through design, purchasing, process control and efficient use.

Reuse

Use an item or component again with little processing, after safety and quality checks.

Recycle

Process segregated material into secondary raw material or products.

Hierarchy: Prevention/reduction is generally preferred over reuse, recycling, recovery and disposal because it avoids upstream resource and energy use.

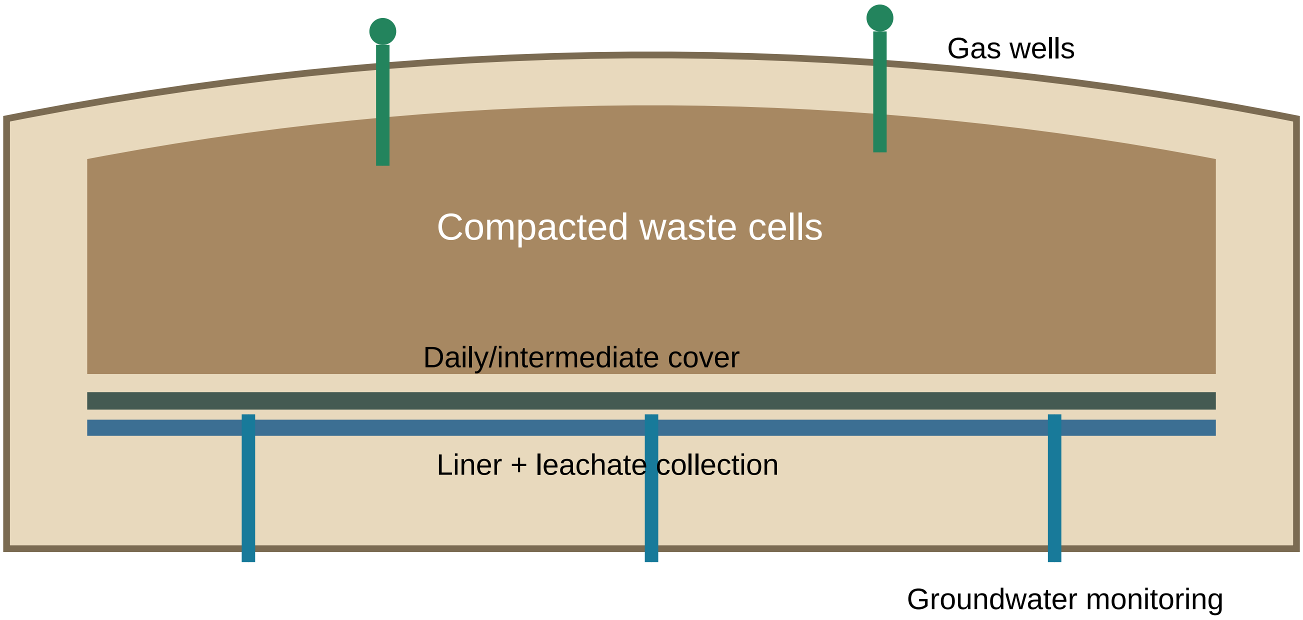
Collection system

1. Characterise waste and generation points.
2. Provide colour/labelled containers suited to the stream.
3. Set collection frequency based on quantity and risk.
4. Use safe internal transport and prevent spillage.
5. Record weight, destination and responsible person.
6. Send each stream only to an authorised next step.

Energy recovery

Suitable high-calorific residual waste may produce energy through controlled thermal or biological processes. Energy recovery does not replace segregation and recycling. Feed quality, air-emission control, ash management, efficiency and life-cycle impact must be evaluated.

Sanitary landfill



An engineered landfill uses liners, leachate collection, compacted cells, cover, gas management, drainage and monitoring.

Hazardous-waste principles

Hazardous waste may be toxic, flammable, corrosive, reactive, infectious or otherwise dangerous. It requires identification, compatible packaging, labels, secondary containment, manifests/records, trained handling and authorised treatment/disposal.

Worked Problem 16

Landfill-volume estimate

Residual waste is 12 tonnes/day. Compacted density is 600 kg/m³. Estimate daily waste volume before cover.

$$\text{Mass} = 12{,}000 \text{ kg/day}$$

$$\text{Volume} = \text{mass/density} = 12{,}000/600 = 20 \text{ m}^3\text{/day}$$

If cover and operational allowance add 15%, planned daily volume ≈ 23 m³/day.

M4.04 • 1 HOUR

Environmental legislation awareness and pollution-control boards

The syllabus lists air-quality and pollution-control acts/rules and asks students to understand the structure and role of central and state pollution-control boards. The exact titles/years shown in teaching documents may contain legacy wording or typographical errors, so real compliance work must verify the current official legal text.

Central-level functions

- National coordination and technical guidance
- Standards, monitoring programmes and data support
- Advising government and supporting laboratories
- Coordination with state boards and committees

State-level functions

- Consent/authorisation functions under applicable law
- Inspection, monitoring and enforcement
- Review of pollution-control systems
- Complaint response, data collection and local implementation

Legal-use caution: This handbook is educational. For a project, consent application or compliance decision, use the current official CPCB/SPCB or government source and obtain competent professional guidance.

What a student should understand

1. Environmental law defines responsibilities, standards, authorities and enforcement mechanisms.
2. Industries may require approvals, monitoring, records and reporting.
3. Boards inspect, sample, issue directions and guide pollution control within their authority.
4. Evidence must be based on approved methods, calibrated instruments and documented records.

M4.05 • 1 HOUR

Carbon footprint and carbon credit

Carbon footprint

A carbon footprint is the total greenhouse-gas emissions associated with an organisation, activity, product or person, expressed as carbon-dioxide equivalent (CO_2e) over a defined boundary and period.

Direct emissions
Fuel burned in owned/controlled equipment.

Purchased-energy emissions
Electricity, steam, heat or cooling purchased.

Other indirect emissions

Materials, transport, waste, travel and supply chain.

$$\text{Emissions} = \text{Activity data} \times \text{Applicable emission factor}$$

Worked Problem 17 **Illustrative monthly footprint**

A workshop uses 200 kWh electricity at an illustrative factor of 0.70 kg CO_2e/kWh and 20 L diesel at 2.31 kg CO_2e/L.

$$\text{Electricity} = 200 \times 0.70 = 140 \text{ kg } CO_2e$$

$$\text{Diesel} = 20 \times 2.31 = 46.2 \text{ kg } CO_2e$$

$$\text{Total} = 186.2 \text{ kg } CO_2e$$

These factors are examples only. Real inventories need current authorised factors, clear boundaries and transparent assumptions.

Carbon credit

A carbon credit generally represents a quantified greenhouse-gas reduction or removal under a defined programme. Credible credits depend on additionality, baseline, measurement, verification, permanence, leakage control and prevention of double counting.

Reduction before offset: First measure emissions, reduce avoidable energy/material use and improve technology. Credits do not replace direct reduction.

M4.06 • 1 HOUR

Environmental management in fabrication industry

Typical aspects and impacts

Activity	Environmental aspect	Potential impact	Control
Cutting/grinding	Metal scrap, dust, noise and energy use	Resource loss, air/noise exposure	Optimised nesting, extraction, enclosure and maintenance
Welding	Fume, gases, electricity and electrodes	Air exposure and resource use	Process selection, local exhaust and housekeeping
Painting	VOC, overspray, containers and sludge	Air pollution and hazardous waste	Low-VOC options, transfer efficiency, booth filters and closed storage
Machining	Coolant, oil, chips and energy	Water/soil pollution and metal loss	Leak prevention, fluid life extension and chip separation
Cleaning	Water, detergent and contaminated effluent	Water use and pollution load	Counter-current cleaning, substitution and treatment
Material handling	Packaging and damaged material	Waste and resource loss	Reusable packaging and proper storage

Aspect–impact evaluation

1. List activities, products and services.
2. Identify environmental aspects under normal, abnormal and emergency conditions.
3. Identify impacts and legal/other obligations.
4. Rank significance using a documented method.
5. Set operational controls, objectives and monitoring.
6. Review incidents, performance and improvement opportunities.

Worked Problem 18 Simple coolant-loss improvement

A workshop purchased 500 L coolant concentrate per year. After leak repair and concentration control, use fell to 350 L with the same production.

$$\text{Reduction} = 500 - 350 = 150\ \text{L/year}$$

$$\text{Percentage reduction} = 150/500 \times 100 = 30\%$$

Benefits may include lower purchase cost, less waste, cleaner floors and reduced environmental risk.

M4.07 • 1 HOUR

ISO 14000 and environmental management systems

ISO 14000 is a family of environmental-management standards. ISO 14001 specifies requirements for an environmental management system (EMS); it does not prescribe one universal emission level for every organisation.

Plan–Do–Check–Act

Plan

Context, leadership, policy, aspects, obligations, risks, opportunities and objectives.

Do

Resources, competence, communication, documentation, operational control and emergency preparedness.

Check

Monitoring, measurement, compliance evaluation, internal audit and management review.
Course 2001

Act

Correct nonconformity and continually improve suitability, adequacy and effectiveness.

Implementation sequence

1. Obtain leadership commitment and define EMS scope.
2. Develop environmental policy.
3. Identify aspects, impacts and compliance obligations.
4. Set measurable objectives and plans.
5. Establish competence, communication and documents.
6. Implement operational and emergency controls.
7. Monitor performance and evaluate compliance.
8. Conduct internal audits and management review.
9. Correct problems and improve continuously.

Benefits

Potential benefit	How it arises
Better control	Clear responsibilities, procedures and records
Lower resource use	Objectives for energy, water and material efficiency
Reduced risk	Aspect assessment and emergency preparedness
Improved compliance management	Systematic tracking and evaluation of obligations
Continual improvement	Audit, review, corrective action and performance monitoring

ISO memory: ISO 14001 certificate മാത്രം ലക്ഷ്യം അല്ല; day-to-day environmental control system പ്രവർത്തിക്കുകയാണ് പ്രധാന്യം.

M4.08 • 3 HOURS

Case-study requirement

The syllabus states that the case study is compulsory for Continuous Internal Evaluation and not for the End Semester Examination. Students may work in groups of 4–5; topics should not be duplicated; each member submits the group report individually.

Course 2001

Suggested direction 1

Engineering solution for a local environmental problem involving pollution, waste, water, noise, energy or material loss.

Suggested direction 2

Study of a sustainable-energy resource with emphasis on a real application, suitability, benefits, limitations and improvement.

Full case-study workflow, sample reports, data tables and evaluation checklist are provided in the Case Study tab.

SERIES TEST II • 1 HOUR

Original practice test — Modules 3 and 4

This is not an official board question paper.

1. Explain the construction and working of a flat-plate collector.
2. Describe anaerobic digestion and biogas utilisation.
3. Calculate simplified wind power for given ρ, A, v and C_p.
4. Compare hydrogen, tidal and geothermal energy.
5. Explain sources and management of e-waste and biomedical waste.
6. Explain 3R and sanitary-landfill components.
7. Define carbon footprint and carbon credit.
8. Prepare an aspect–impact table for a fabrication workshop.
9. Explain PDCA in an ISO 14001 environmental management system.
10. Prepare a case-study outline for one local environmental problem.

Environmental Case Study Handbook

A practical workflow for groups of 4–5 to investigate a real problem or sustainable-energy application and submit individual copies of the group report.

Case-study workflow

1. **Select and approve the topic:** choose one local, observable problem or energy application that does not duplicate another group.
2. **Define the boundary:** location, activity, period, people/equipment and environmental media included.

3. **Write objectives:** specify what will be measured, compared or improved.
4. **Collect baseline evidence:** observation, count, weight, meter reading, interview, photograph, map or authorised record.
5. **Analyse source–pathway–receptor:** identify causes, transport routes and affected receptors.
6. **Generate alternatives:** prevention, reduction, reuse, control, treatment and monitoring options.
7. **Evaluate alternatives:** technical feasibility, environmental benefit, cost, safety, maintenance and acceptance.
8. **Recommend a solution:** include equipment/process, responsibilities, timeline and performance indicators.
9. **Prepare report:** use tables, diagrams, calculations, limitations, conclusion and references.
10. **Present and defend:** every group member must explain data, calculation and recommendation.

Ethics and safety: Do not enter restricted areas, touch hazardous waste, collect biological samples, publish personal data or operate equipment without permission and supervision.

Required report structure

Chapter	Required content
Cover page	Course, title, institution, group, individual student and date
Certificate/declaration	Prescribed text and signatures where required
Abstract	Problem, method, major finding and recommendation in one page
Introduction	Background, significance and study location
Objectives and scope	Clear questions, boundaries and limitations
Method	Observation, measurement, sampling or data-collection procedure
Results	Tables, figures, photographs and calculations
Discussion	Interpretation, comparison and causes
Engineering solution	Alternatives, selection criteria, design concept and controls
Implementation plan	Responsibilities, timeline, resources and indicators
Conclusion	Direct answer to objectives
References and annexures	Sources, raw data, questionnaire and additional evidence

Sample Case Study A — Campus biodegradable waste

Problem statement

Mixed food waste from a campus canteen causes odour, attracts animals and contaminates recyclables. The study evaluates source segregation and a small composting/biogas option.

Objectives

1. Estimate daily biodegradable waste.
2. Identify current handling gaps.
3. Compare composting and anaerobic digestion.
4. Recommend a safe, maintainable system.

Seven-day demonstration dataset

Day	Food preparation waste (kg)	Plate waste (kg)	Total (kg)
1	18	12	30
2	17	14	31
3	20	13	33
4	19	11	30
5	21	15	36
6	16	10	26
7	15	9	24

$$\text{Weekly total} = 210\ \text{kg; average} = 210/7 = 30\ \text{kg/day}$$

If 80% is suitable for biological treatment after segregation, design basis = 24 kg/day. Add a safety margin only after observing seasonal and event variation.

Alternatives

Option	Strength	Limitation
Aerobic composting	Simple, produces soil amendment	Requires bulking, aeration, moisture and odour control
Anaerobic digestion	Produces biogas and digestate	Higher process/safety skill and stable feed needed
Authorised external treatment	Lower on-site operation	Transport, service reliability and recurring cost

Recommendation

Begin with two-bin segregation, weighing and contamination monitoring. Select composting or digestion only after space, operator skill, odour, water, digestate/compost use and approval requirements are verified.

Performance indicators

- ✓ kg/day segregated
- ✓ contamination percentage
- ✓ odour/complaint count
- ✓ product quantity and quality
- ✓ downtime and maintenance
- ✓ residual waste sent for disposal

Sample Case Study B — Solar hot water for a hostel

Baseline

A hostel needs 1,000 L/day of water raised from 25°C to 55°C.

$$\text{Useful heat} = 1{,}000\ \text{kg} \times 4.18\ \text{kJ/kg}\cdot{}^\circ\text{C} \times 30^\circ\text{C} = 125{,}400\ \text{kJ/day} = 125.4\ \text{MJ/day}$$

$$\text{Electrical equivalent} = 125.4/3.6 = 34.83\ \text{kWh/day}$$

If an existing electric heater system uses about 38 kWh/day including losses, annual energy for 300 operating days is 11,400 kWh.

Study tasks

1. Measure actual hot-water quantity and temperature profile.
2. Check roof area, orientation, shading, structure and pipe route.
3. Estimate solar resource and system losses using approved design data.
4. Compare flat-plate and evacuated-tube systems where relevant.
5. Plan storage, backup, scaling control, insulation and maintenance.
6. Estimate energy and emissions avoided using current factors.
7. Assess life-cycle cost, safety and user behaviour.

> **Conclusion format:** State technical suitability, expected useful-energy contribution, required collector/storage arrangement, controls, uncertainty and monitoring plan.

Case-study submission checklist

Evidence

- ✓ Dated observations
- ✓ Raw data table
- ✓ Site map/photo permission
- ✓ Calculation units
- ✓ Source references

Analysis

- ✓ Source–pathway–receptor
- ✓ Cause analysis
- ✓ At least two alternatives
- ✓ Safety and legal awareness
- ✓ Limitations

Recommendation

- ✓ Responsible person
- ✓ Timeline
- ✓ Resource need
- ✓ Monitoring indicators
- ✓ Expected benefit

Practice Questions and Environmental Problem Laboratory

Searchable short answers, diagrams, engineering scenarios and numerical practice.

Module 1 practice

1. Draw an ecosystem showing energy and nutrient flow.
2. Compare pyramid of number, biomass and energy.
3. Create a food web with at least eight organisms.
4. Explain lentic and lotic systems using local examples.
5. Draw carbon and nitrogen cycles.
6. Explain how fertiliser changes nitrogen/phosphorus cycles.
7. Differentiate greenhouse effect and global warming.
8. Explain ozone depletion mechanism and effects.
9. Trace acid-rain formation from fuel combustion.
10. Calculate energy at four trophic levels using an assumed transfer percentage.

Module 2 practice

1. Apply source–pathway–receptor to a roadside workshop.
2. Classify ten pollutants as particulate/gaseous and primary/secondary.
3. Draw cyclone, bag filter and ESP.
4. Select control equipment for cement dust, boiler dust and soluble gas.
5. Calculate outlet concentration for 95% removal.

6. Compare pH, DO, BOD and COD.
7. Interpret a river sample with DO 3 mg/L and BOD 20 mg/L.
8. Calculate BOD reduction from 250 to 25 mg/L.
9. Prepare source-path-receiver controls for noise.
10. Combine two equal 90 dB sources.

Module 3 practice

1. Label a flat-plate collector.
2. Compare liquid and air collectors.
3. Explain solar pond, dryer and still.
4. Calculate heat for 200 L water through 40°C rise.
5. Explain biomass fuel properties.
6. Draw anaerobic-digestion stages.
7. Estimate biogas from a given daily feed.
8. Calculate wind power using given data.
9. Compare wind benefits and problems.
10. Compare hydrogen, wave, tidal, OTEC and geothermal systems.

Module 4 practice

1. Characterise municipal, electronic and biomedical waste.
2. Prepare a segregation plan for a campus.
3. Develop a material balance for a workshop.
4. Explain 3R with engineering examples.
5. Draw a sanitary-landfill cross-section.
6. Calculate landfill volume from mass and compacted density.
7. Prepare a simple carbon-footprint inventory.
8. Create an aspect–impact register for welding and painting.
9. Explain PDCA for ISO 14001.
10. Prepare a complete case-study proposal.

Challenge problems with hints

1. Ecological energy

A producer level has 50,000 kJ and transfer is 12% per trophic level. Find energy at three consumer levels. **Hint:** multiply by 0.12 repeatedly.

2. Air-control efficiency

Inlet dust is 800 mg/m³ and outlet 32 mg/m³. Find efficiency. **Hint:** (inlet − outlet)/inlet × 100.

3. Water treatment

BOD falls from 320 to 48 mg/L. Find removal and discuss whether one value proves compliance. **Hint:** calculate 85%; compliance also needs current standard and method.

4. Solar heat

Heat 250 kg water from 28°C to 62°C. **Hint:** $Q = mc\Delta T$.

5. Wind speed change

All else equal, compare power at 6 and 9 m/s. **Hint:** ratio = $(9/6)^3 = 3.375$.

6. Waste diversion

Waste = 500 kg/day; 45% composted and 30% recycled. Find disposal fraction. **Hint:** 25% or 125 kg/day.

7. Carbon reduction

Electricity decreases from 1,200 to 900 kWh/month at 0.65 kg CO_2e/kWh. **Hint:** 300×0.65.

Common mistakes and corrections

Mistake	Correction
Calling every environmental change “pollution”	Identify adverse change, pollutant, concentration/duration and receptor.
Arrow reversed in food chain	Arrow follows energy transfer from food to consumer.
Saying energy cycles	Energy flows and dissipates; matter cycles.
Confusing ozone depletion with global warming	Explain separate gases, mechanisms and effects.
Assuming COD and BOD are identical	State the different test principles and what each indicates.
Adding decibels arithmetically	Use logarithmic addition.
Quoting renewable energy as impact-free	Discuss life-cycle, siting, material and operational impacts.
Treating mixed waste as recyclable	Segregate clean streams at source.
Using outdated law names without verification	Check current official text and amendments.
Giving a case-study solution without baseline data	Measure/observe first, then compare alternatives.

Quick Revision Map

Compact memory cues for all four modules.

Ecosystem

Abiotic + biotic interactions.

Energy

Flows; nutrients cycle.

Food web

Interconnected food chains.

Lentic
Standing water.

Lotic
Flowing water.

Carbon cycle
Photosynthesis, respiration, decomposition, combustion.

Nitrogen
Fixation → nitrification → assimilation → ammonification → denitrification.

Global warming
Enhanced greenhouse effect.

Acid rain
Mainly SO_2/NO_x transformation.

Pollution model
Source → pathway → receptor.

Cyclone
Centrifugal separation.

Bag filter
Fabric filtration.

ESP
Electrical charging/collection.

DO
Dissolved oxygen.

BOD
Biological oxygen demand.

COD
Chemical oxygen demand.

Noise
Logarithmic dB scale.

Solar collector

Cover + absorber + tubes + insulation.

Biogas

Four anaerobic stages.

Wind

$P \propto v^3$.

Hydrogen

Energy carrier.

3R

Reduce, reuse, recycle.

Landfill

Liner, leachate, gas, cover, monitoring.

Carbon footprint

Activity × factor.

EMS

Plan–Do–Check–Act.

Case study

Baseline → alternatives → recommendation → indicators.

Revision order: Ecosystem → cycles → atmospheric problems → pollution devices → water/noise parameters → renewable systems → waste hierarchy → carbon/EMS → case-study method.

Forty Important Questions and Answers

1. What is an ecosystem?

A community of organisms interacting with the physical and chemical environment.

2. Producer?

An organism that makes organic matter from inorganic materials using an energy source.

3. Decomposer?

An organism that breaks down dead matter and returns nutrients.

4. Food web?

A network of interconnected food chains.

5. Why is energy pyramid upright?

Energy is lost as heat at every trophic transfer.

6. Lentic versus lotic?

Lentic is standing water; lotic is flowing water.

7. Nitrogen fixation?

Conversion of atmospheric nitrogen into biologically usable compounds.

8. Eutrophication?

Nutrient enrichment causing excessive growth and possible oxygen depletion.

9. Greenhouse effect?

Atmospheric absorption and re-emission of outgoing infrared energy.

10. Acid rain precursor?

Mainly sulphur dioxide and nitrogen oxides.

11. Pollutant?

A substance or energy form capable of adverse effect at a given concentration/duration.

12. Primary pollutant?

Emitted directly from a source.

13. Cyclone principle?

Centrifugal separation of particles from gas.

14. Bag-filter principle?

Particles are captured as gas passes through fabric.

15. ESP principle?

Charged particles migrate to collection plates.

16. Absorber?

A device transferring a gas pollutant into a liquid.

17. Catalytic converter?

A vehicle exhaust device catalysing reactions that reduce CO, hydrocarbons and NO_x.

18. DO?

Dissolved oxygen available in water.

19. BOD?

Oxygen used biologically to degrade biodegradable matter under test conditions.

20. COD?

Oxygen-equivalent measure of chemically oxidisable material under the test.

21. Why not add dB directly?

The decibel scale is logarithmic.

22. Flat-plate absorber?

A dark surface converting solar radiation to heat.
Course 2001

23. Selective coating?

High solar absorption with relatively low thermal emission.

24. Four digestion stages?

Hydrolysis, acidogenesis, acetogenesis and methanogenesis.

25. Main biogas component?

Methane, with carbon dioxide and minor constituents.

26. Wind power relation?

Available power is proportional to air density, swept area and wind speed cubed.

27. Hydrogen source or carrier?

Generally an energy carrier.

28. Tidal energy advantage?

Tides are highly predictable.

29. Geothermal binary cycle?

Geothermal fluid heats a separate low-boiling working fluid.

30. E-waste?

Discarded electrical/electronic equipment and components.

31. 3R?

Reduce, reuse and recycle.

32. Leachate?

Contaminated liquid draining through waste.

33. Hazardous waste?

Waste with dangerous properties such as toxicity, corrosivity, reactivity or flammability.

34. Carbon footprint?

Total greenhouse-gas emissions within a defined boundary, expressed as CO_2e.

35. Carbon credit?

A quantified reduction/removal unit under a defined programme.

36. Environmental aspect?

An element of an activity, product or service that interacts with the environment.

37. Environmental impact?

A change to the environment resulting from an aspect.

38. ISO 14001?

A standard specifying requirements for an environmental management system.

39. PDCA?

Plan, Do, Check and Act.

40. First case-study step?

Define an approved problem, objective, boundary and safe data method.

Series-Test Answers, Solved Challenge Problems and References

Series Test I answer guidance

1. **Ecosystem:** define community–environment interaction; list biotic producers/consumers/decomposers and abiotic water, air, soil, light and temperature.
2. **Food chain/web/pyramid:** give definitions, arrow direction, one example and explain energy loss.
3. **Cycle:** draw reservoirs and labelled processes; carbon or nitrogen earns strong coverage.
4. **Atmospheric problems:** explain different mechanism, gases, effects and controls.
5. **Pollution:** define pollutant and show source–pathway–receptor.
6. **Control device:** include construction, working principle, application, advantage and limitation.
7. **DO/BOD/COD:** distinguish oxygen present, biological demand and chemical demand.
8. **BOD removal:** (240 − 36)/240 × 100 = 85%.
9. **Noise:** sources, effects, source/path/receiver controls and regulatory awareness.
10. **Two 85 dB sources:** approximately 88 dB.

Series Test II answer guidance

1. **Flat plate:** transparent cover, absorber/coating, flow passage, insulation, casing and heat-transfer process.
2. **Digestion:** explain four stages, gas composition, uses, digestate and safety.
3. **Wind numerical:** use $\frac{1}{2}\rho A v^3 C_p$ with units and state simplifications.
4. **New sources:** compare origin, conversion, application and challenge.
5. **E/biomedical waste:** sources, characteristics, risk, segregation and authorised management.
6. **3R/landfill:** hierarchy plus liner, leachate, gas, cover and monitoring.
7. **Carbon:** footprint is inventory; credit is verified reduction/removal unit.
8. **Fabrication:** include activity, aspect, impact, control and indicator.
9. **ISO:** Plan–Do–Check–Act with aspects, objectives, controls, audit and improvement.
10. **Case study:** problem, baseline, method, alternatives, recommendation and indicators.

Challenge-problem answers

Problem	Answer
Ecological energy	50,000 × .12 = 6,000 kJ; × .12 = 720 kJ; × .12 = 86.4 kJ.
Air-control efficiency	(800 − 32)/800 × 100 = 96%.
BOD removal	(320 − 48)/320 × 100 = 85%; compliance requires current standard and method.
Solar heat	Q = 250 × 4.18 × 34 = 35,530 kJ = 35.53 MJ.
Wind ratio	$(9/6)^3$ = 3.375; theoretical available power is 3.375 times.
Waste diversion	75% diverted; 25% disposed = 125 kg/day.
Carbon reduction	(1,200 − 900) × 0.65 = 195 kg CO_2e/month.

Text and reference list from the syllabus

1. C. N. R. Rao — *Understanding Chemistry*, Universities Press, 2011.
2. William Nazaroff and Lisa Cohen — *Environmental Engineering Science*.
3. C. S. Rao — *Environmental Pollution Control and Engineering*.
4. M. N. Rao and H. V. N. Rao — *Air Pollution*.
5. Frank Kreith and Jan F. Kreider — *Principles of Solar Engineering*.
6. Aldo Vieira da Rosa — *Fundamentals of Renewable Energy Processes*.
7. A. D. Patwardhan — *Industrial Solid Waste*.
8. Metcalf & Eddy — *Wastewater Engineering*.

The syllabus also lists environmental portals including TERI, pollution-control resources, India Environment Portal, United Nations Sustainable Development Goals and general energy-conservation resources. Verify current official web addresses before use because legacy syllabus URLs may contain typographical or domain changes.

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